Week 11
Contextual Bandits and Off-Policy Evaluation

STUDY-RL — a self-paced Deep RL curriculum

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Learning objectives

Study companion. Read `teaching_guide.md` in this folder for step-by-step examples, an analogy with its limits, and self-checks. Then use `workbook.ipynb` to implement and vary the calculations.

Key Idea

By the end of this lecture you will be able to:

1. Derive the UCB1 regret bound sketch from Hoeffding and Lai–Robbins, and contrast it with KL-UCB, MOSS, and Thompson Sampling.
2. Show that the LinUCB posterior mean is the closed-form ridge solution $\hat{\theta}_a = A_a^{-1} b_a$ with $A_a = \lambda I + \sum_t x_t x_t^\top$.
3. Prove that Inverse Propensity Scoring (IPS) is unbiased, derive its variance, and explain why Self-Normalised IPS (SNIPS) is biased but often dominant in mean squared error.
4. Derive the Doubly Robust estimator and prove its double-robustness: consistent if *either* the propensity *or* the reward model is correct.
5. Build a synthetic insurance lead–broker contextual bandit (6 brokers, 8 lead features), collect ε-greedy logs, and run SNIPS, Direct, DR, and SWITCH-DR against three counterfactual policies.
6. Read off a bias/variance trade-off plot and pick τ for SWITCH.
7. State the four production gotchas: logging propensities, overlap (positivity) failure, delayed feedback bias, and propensity overfitting.

1 Motivation and intuition

A *bandit* is a Markov Decision Process collapsed to one step: at every round you observe a context x_t, pick an arm $a_t \in \{1, \ldots, K\}$, and receive a noisy reward r_t. There is no transition; the next round's context is independent of your action.

The puzzle is the same as in MDPs — balance *exploration* (try arms whose value you do not know) with *exploitation* (use the best arm you have seen). What is gone is the credit-assignment chain. What is gained is **a fully solvable, tightly characterised problem** where every algorithmic trick (UCB, posterior sampling, importance sampling, doubly robust estimation) has a clean derivation and a sharp regret bound. Master this and the deeper RL material reduces to "do this, but with a target network and a replay buffer".

Why bother with such a small problem? Because almost every applied ML system is at heart a bandit once the deep-learning fog clears: pick this ad / pick this thumbnail / pick this broker.

1.1 Two running examples

synthetic insurance lead-broker matching. A motor-insurance comparator routes each incoming *lead* (a quote form) to one of K brokers. The reward is *bind*: did the broker convert the lead into a sold policy? Features include postcode band, vehicle power, prior claims, channel (direct vs. aggregator), age band. The platform must (i) maximise binds, (ii) keep brokers happy by sending them leads that match their book of business, and (iii) audit the policy offline before pushing changes to production. *This is a contextual bandit with delayed, partially observed feedback.*

ATS candidate–job ranking. (Treated in depth in Week 12.) A recruiter platform must rank candidates for a requisition. The reward is hire. Position bias makes naive click-modelling actively misleading. The re-ranker is a contextual bandit on top of a two-tower retriever.

1.2 Anchoring to what you already know

LinUCB uses a linear reward model with a confidence bonus. Its ridge estimate is not logistic regression; a sigmoid conditional mean is a model mismatch unless the method is changed. Thompson sampling draws from a posterior or an exploration distribution and selects an action. Off-policy evaluation resembles held-out evaluation, but selected-action feedback, support and logging probabilities add identification issues that ordinary cross-validation does not solve.

Common pitfall

Bandit literature is split between the *stochastic* setting (rewards i.i.d. given arm) and the *adversarial* setting (rewards chosen by an adversary). Almost everything in this chapter is stochastic. Production matching is somewhere in between: brokers and recruiters are slow-moving adversaries (incentive-compatibility) on top of i.i.d. leads. We address that in Week 12; for now, assume i.i.d.

2 Mathematical formulation

Definition 2.0.1 (Stochastic K-armed bandit)**.** A tuple $(\mathcal{A}, P)$ with $|\mathcal{A}| = K$ and $P = (P_a)_{a\in\mathcal{A}}$ a family of reward distributions on $[0,1]$. At each round $t = 1, 2, \dots$ the learner picks $A_t \in \mathcal{A}$ and observes $R_t \sim P_{A_t}$ independently of the past given A_t. Let $\mu_a = \mathbb{E}[R \mid a]$ and $\mu^\star = \max_a \mu_a$. The *regret* is

$$\mathrm{Reg}(T) \triangleq T\mu^\star - \mathbb{E}\Big[\sum_{t=1}^{T} R_t\Big] = \sum_a \Delta_a\, \mathbb{E}[N_a(T)],$$

where $\Delta_a = \mu^\star - \mu_a$ and $N_a(T) = \sum_t \mathbb{1}\{A_t = a\}$.

Theorem 2.0.2 (Lai–Robbins lower bound, 1985)**.** *For any* consistent *policy and any Bernoulli bandit,* $\liminf_{T\to\infty} \mathrm{Reg}(T)/\log T \geq \sum_{a:\Delta_a>0} \Delta_a/\mathrm{KL}(\mu_a, \mu^\star)$.

The lower bound says: regret must grow at least like $\log T$, and the "hardness" constant is governed by KL divergences. UCB1 achieves the $\log T$ rate with regret coefficient proportional to $1/\Delta_a$; its pull-count coefficient is proportional to $1/\Delta_a^2$; KL-UCB achieves the KL constant.

2.1 UCB1 and Hoeffding

Theorem 2.1.1 (Hoeffding's inequality)**.** *Let $X_1, \dots, X_n$ be i.i.d. in $[0,1]$ with mean μ. Then for any $\delta > 0$,*

$$\mathbb{P}\Big(|\bar{X}_n - \mu| > \sqrt{\tfrac{\log(2/\delta)}{2n}}\Big) \leq \delta.$$

UCB1 (Auer, Cesa-Bianchi, Fischer 2002) plays

$$A_t = \arg\max_a \Big[\hat{\mu}_a(t-1) + \sqrt{\tfrac{2\log t}{N_a(t-1)}}\Big].$$

The bonus is the Hoeffding radius with $\delta = 2t^{-4}$ — shrinking fast enough that the bad-confidence event is summable.

Theorem 2.1.2 (UCB1 regret, sketch)**.** $\mathbb{E}[\mathrm{Reg}(T)] \leq \sum_{a:\Delta_a>0} \big(\frac{8\log T}{\Delta_a} + (1+\pi^2/3)\Delta_a\big)$.

Sketch. Pull each arm once. Fix a suboptimal arm and $\ell = \lceil 8\log T/\Delta_a^2 \rceil$. Once $N_a \geq \ell$, selection requires an optimistic deviation for that arm or a pessimistic deviation for the best arm. Hoeffding applies to fixed sample counts; union-bound over both possible counts up to t to obtain at most $2t^2 t^{-4} = 2t^{-2}$. Thus $\mathbb{E}[N_a(T)] \leq 8\log T/\Delta_a^2 + 1 + \pi^2/3$. Multiply by Δ_a and sum. A small-count condition is $N_a \leq 8\log t/\Delta_a^2$, not the reversed inequality. □

Remark. The slack between Lai–Robbins (Δ_a/KL) and UCB1 ($\Delta_a/\Delta_a^2 = 1/\Delta_a$) comes from Hoeffding being loose for Bernoulli rewards near 0 or 1. KL-UCB closes this slack by replacing Hoeffding with the *KL* upper confidence bound:

$$\mathrm{UB}_a(t) = \max\{q \in [\hat{\mu}_a, 1] : N_a \cdot \mathrm{KL}(\hat{\mu}_a, q) \leq \log t + c\log\log t\}.$$

2.2 Thompson Sampling

Definition 2.2.1 (Thompson Sampling, Bernoulli)**.** Maintain a Beta(α_a, β_a) posterior per arm. Initialise $\alpha_a = \beta_a = 1$. At round t: sample $\tilde{\mu}_a \sim \text{Beta}(\alpha_a, \beta_a)$, play $A_t = \arg\max_a \tilde{\mu}_a$, observe $R_t \in \{0, 1\}$, update $\alpha_{A_t} \mathrel{+}= R_t$, $\beta_{A_t} \mathrel{+}= 1 - R_t$.

Theorem 2.2.2 (Agrawal & Goyal 2012/2013)**.** *For stationary independent Bernoulli rewards, the 2013 analysis gives $O(\sum_{a:\Delta_a>0}(\log T)/\Delta_a)$ plus lower-order instance-dependent terms. The original 2012 general-arm bound had looser gap dependence; a rate statement alone does not establish the exact Lai–Robbins coefficient. The proof reduces to bounding the time the posterior on a sub-optimal arm dominates the optimal arm's posterior, using anti-concentration of the Beta.*

Intuition. TS is "OFU by sampling". UCB picks the upper confidence bound; TS picks a draw from the posterior. Both are aggressive enough to explore, but TS is *randomised*, which makes it composable with batched updates, delayed feedback, and ensembles. Compare TS and UCB on the task and uncertainty model at hand; posterior sampling is not a universal empirical winner.

2.3 Contextual bandits and LinUCB

Definition 2.3.1 (Linear contextual bandit)**.** At each round t: observe $x_t \in \mathbb{R}^d$ with $\|x_t\|_2 \le 1$; pick $A_t \in [K]$; observe $R_t = \langle x_t, \theta_{A_t}\rangle + \eta_t$, with η_t conditionally σ-sub-Gaussian. Hidden parameters θ_a are bounded ($\|\theta_a\|_2 \le S$).

Theorem 2.3.2 (Ridge estimate and its Gaussian interpretation)**.** *Let $A_a = \lambda I + X_a^T X_a$ and $b_a = X_a^T r_a$, with $\lambda > 0$. Minimising $\|r_a - X_a\theta\|^2 + \lambda\|\theta\|^2$ gives $\hat{\theta}_a = A_a^{-1} b_a$. With Gaussian noise variance σ^2 and prior covariance $(\sigma^2/\lambda)I$, the posterior is $\mathcal{N}(\hat{\theta}_a, \sigma^2 A_a^{-1})$. A prior covariance $\lambda^{-1}I$ instead changes the ridge term to $\sigma^2\lambda I$.*

Proof. Differentiate the quadratic, set $2(A_a\theta - b_a) = 0$ and complete the same square in the Gaussian log posterior. Ridge estimation remains a well-defined procedure outside Gaussian noise; that does not make its uncertainty exactly Bayesian under every reward model. □

LinUCB acts by

$$A_t = \arg\max_a \langle x_t, \hat{\theta}_a\rangle + \alpha\sqrt{x_t^\top A_a^{-1} x_t},$$

where the bonus is a directional confidence width. Under bounded features, conditionally σ-sub-Gaussian noise and $\|\theta_a\| \le S$, one valid self-normalised radius for arm a is

$$\alpha_{a,t} = \sigma\sqrt{2\log\left(\frac{\det(A_{a,t})^{1/2}}{\det(\lambda I)^{1/2}(\delta/K)}\right)} + \sqrt{\lambda}S.$$

The union over K arms allocates failure probability δ/K. A fixed empirical alpha in the project is not this complete confidence schedule.

Remark (Shared and disjoint models have different dimensions). A shared d-parameter linear bandit has the familiar $\widetilde{O}(d\sqrt{T})$ regret bound under its assumptions. The displayed disjoint-arm model contains K separate d-vectors. Summing elliptical potentials across arms and applying Cauchy–Schwarz gives a typical $\widetilde{O}(d\sqrt{KT})$ bound (with regularity and confidence choices), not a bound independent of K. For example, with $\lambda \ge 1$ and $\|x\| \le 1$, $\sum_t x_t^T A_{A_t,t-1}^{-1} x_t \le 2Kd\log(1 + T/(Kd\lambda))$. Source: https://papers.nips.cc/paper/2011/hash/e1d5be1c7f2f456670de3d53c7b54f4a-Abstract.html.

2.4 MOSS: removing the spurious $\log T$

UCB1 has a $\log T$ term in the bonus that produces a $\log T$ factor in the worst-case (minimax) regret. *MOSS* (Audibert & Bubeck 2009) replaces the bonus with $\sqrt{\log_+(T/(Kn_a))/n_a}$, where $\log_+(z) = \max(0, \log z)$. The minimax regret becomes $O(\sqrt{KT})$ — this is the minimax-optimal rate for K-armed bandits with horizon T.

Intuition. For fixed positive reward gaps, logarithmic regret can be appropriate; worst-case bounds allow the gaps to shrink with the horizon. UCB1's usual gap-free upper bound has an extra square-root logarithmic factor, whereas MOSS achieves $O(\sqrt{KT})$. Neither this minimax comparison nor one Bernoulli experiment establishes a universal empirical ordering of UCB1, KL-UCB, MOSS and Thompson sampling.

2.5 LinTS and neural extensions

LinTS (Agrawal & Goyal 2013) replaces the UCB radius with one Gaussian draw per arm: $\tilde{\theta}_a \sim \mathcal{N}(\hat{\theta}_a, v^2 A_a^{-1})$, then act greedily. The cited $\widetilde{O}(d^{3/2}\sqrt{T})$ frequentist result concerns a shared d-parameter payoff model and its prescribed exploration scale. It cannot be transferred unchanged to K unrelated per-arm parameter vectors.

NeuralUCB, NeuralTS, bootstrapped TS. When the reward is non-linear in features (CV embeddings vs. JD embeddings), the linear posterior is inappropriate. The pragmatic stack is: a frozen text encoder produces features $\phi(x) \in \mathbb{R}^d$, then a *Bayesian linear head* per arm runs LinUCB or LinTS. The encoder is shared (no per-arm capacity blow-up), the head still needs calibration and model-fit checks. Week 12 uses a synthetic teaching version rather than a production deployment.

2.6 Off-Policy Evaluation: setup

You have a log $\mathcal{D} = \{(x_i, a_i, r_i, p_i)\}_{i=1}^n$ with $p_i = \pi_b(a_i \mid x_i)$ the *recorded propensity* of the behaviour (logging) policy. You want to estimate

$$V(\pi) = \mathbb{E}_{x\sim D_x}\, \mathbb{E}_{a\sim\pi(\cdot|x)}\, \mathbb{E}_{r\sim P(\cdot|x,a)}[r]$$

for a counterfactual target π *without* deploying π.

Definition 2.6.1 (IPS estimator, Horvitz–Thompson)**.**

$$\hat{V}_{\mathrm{IPS}}(\pi) = \frac{1}{n}\sum_{i=1}^{n} \frac{\pi(a_i \mid x_i)}{\pi_b(a_i \mid x_i)} r_i.$$

Theorem 2.6.2 (IPS is unbiased under positivity)**.** *Assume correct logged propensities, a fixed evaluation policy independent of the evaluation sample, the same context distribution and conditional reward mechanism at logging and evaluation, and* positivity*: $\pi_b(a \mid x) > 0$ whenever $\pi(a \mid x) > 0$. Condition on all outcome-relevant action-selection variables (or randomise actions). Then $\mathbb{E}[\hat{V}_{\mathrm{IPS}}(\pi)] = V(\pi)$.*

Proof. For a single sample,

$$\begin{aligned}
\mathbb{E}\Big[\tfrac{\pi(a|x)}{\pi_b(a|x)} r\Big] &= \mathbb{E}_x \mathbb{E}_{a\sim\pi_b(\cdot|x)} \mathbb{E}_{r\sim P(\cdot|x,a)}\Big[\tfrac{\pi(a|x)}{\pi_b(a|x)} r\Big] \\
&= \mathbb{E}_x \sum_a \pi_b(a|x) \cdot \tfrac{\pi(a|x)}{\pi_b(a|x)} \cdot \mathbb{E}_r[r|x,a] \\
&= \mathbb{E}_x \sum_a \pi(a|x)\, q(x,a) = V(\pi).
\end{aligned}$$

□ □

Theorem 2.6.3 (IPS variance)**.**

$$\mathrm{Var}[\hat{V}_{\mathrm{IPS}}(\pi)] = \frac{1}{n}\Big(\mathbb{E}\Big[\tfrac{\pi(a|x)^2}{\pi_b(a|x)^2} r^2\Big] - V(\pi)^2\Big),$$

For i.i.d. logs and finite second moments, this is governed by $\mathbb{E}_b[\rho^2 r^2]$, not $\mathbb{E}_b[\rho]$, which equals 1 under support. Peaky targets can produce large variance when logger probability is small on their preferred actions; a deterministic target with good overlap need not.

Definition 2.6.4 (SNIPS)**.**

$$\hat{V}_{\mathrm{SNIPS}}(\pi) = \frac{\sum_i w_i r_i}{\sum_i w_i}, \quad w_i = \tfrac{\pi(a_i|x_i)}{\pi_b(a_i|x_i)}.$$

SNIPS is generally biased in finite samples and consistent under support and integrability. An $O(1/n)$ bias expansion needs suitable moment conditions. Its bias sign and variance ordering relative to IPS are not universal; a zero denominator is undefined.

Definition 2.6.5 (Direct method)**.** Fit a reward model $\hat{q}(x, a) \approx \mathbb{E}[r \mid x, a]$ on the log, then

$$\hat{V}_{\mathrm{DM}}(\pi) = \frac{1}{n} \sum_i \sum_a \pi(a \mid x_i)\, \hat{q}(x_i, a).$$

Low variance, biased by model misspecification.

Theorem 2.6.6 (Doubly Robust, Dudík–Langford–Li 2011)**.**

$$\hat{V}_{\mathrm{DR}}(\pi) = \frac{1}{n} \sum_i \Big[\sum_a \pi(a|x_i)\hat{q}(x_i, a) + w_i(r_i - \hat{q}(x_i, a_i))\Big].$$

With support, a fixed target policy, and nuisance predictions independent of each evaluation row, correctness of either the propensity or conditional reward model gives unbiasedness. Use sample splitting or cross-fitting for learned nuisance models; same-row fitting does not automatically retain the exact finite-sample result.

Proof. **Case 1: π_b correct.** Take expectation conditional on x: $\mathbb{E}[w(r - \hat{q}(x, a)) \mid x] = \sum_a \pi(a|x)(q(x, a) - \hat{q}(x, a))$. Adding $\sum_a \pi(a|x)\hat{q}(x, a)$ gives $\sum_a \pi(a|x)q(x, a) = V(\pi|x)$.

Case 2: $\hat{q} = q$. Then $r - \hat{q}(x, a)$ is mean-zero given x, a, so the second term is mean-zero regardless of π_b. The first term equals $V(\pi|x)$. □ □

Remark (DR variance). $\mathrm{Var}[\hat{V}_{\mathrm{DR}}] - \mathrm{Var}[\hat{V}_{\mathrm{IPS}}]$ is governed by weighted reward residuals and variation in the model-based conditional value. A useful reward model often reduces variance, but arbitrary approximation changes do not give a universal monotonic comparison. In the best case (perfect $\hat{q}$), DR still includes weighted outcome noise. It reduces to direct-method variance only when reward residuals have zero conditional variance; knowing a Bernoulli mean does not remove Bernoulli noise.

Definition 2.6.7 (SWITCH-DR, Wang–Agarwal–Dudík 2017)**.** Pick a threshold τ. For each sample with weight $w_i \le \tau$ use DR; for $w_i > \tau$ fall back to the direct method. Trades a small bias for a big variance cut.

2.7 MIPS: OPE for large action spaces

When $K \gg n$ (think: $K = 6000$ jobs, $n = 50000$ logs) IPS variance is fatal because most (x, a) pairs are never observed. *MIPS* (Marginalised IPS; Saito & Joachims, ICML 2022) embeds the action into $e \in \mathbb{R}^{d_e}$ and replaces the per-action propensity with the marginal action-embedding propensity:

$$\hat{V}_{\mathrm{MIPS}}(\pi) = \frac{1}{n} \sum_i \frac{\sum_{a'} \pi(a' \mid x_i) p(e_i \mid x_i, a')}{\sum_{a'} \pi_b(a' \mid x_i) p(e_i \mid x_i, a')} r_i.$$

Theorem 2.7.1 (MIPS identification conditions)**.** *Under embedding sufficiency ($r \perp\!\!\!\perp a \mid x, e$), correct marginal embedding weights, and target-to-logger embedding support, MIPS is unbiased. Marginalisation can reduce variance, but no general $O(d_e/n)$ versus $O(K/n)$ comparison follows from embedding dimension alone. See Saito and Joachims (2022), assumptions 3.1–3.2.*

2.8 Effective sample size

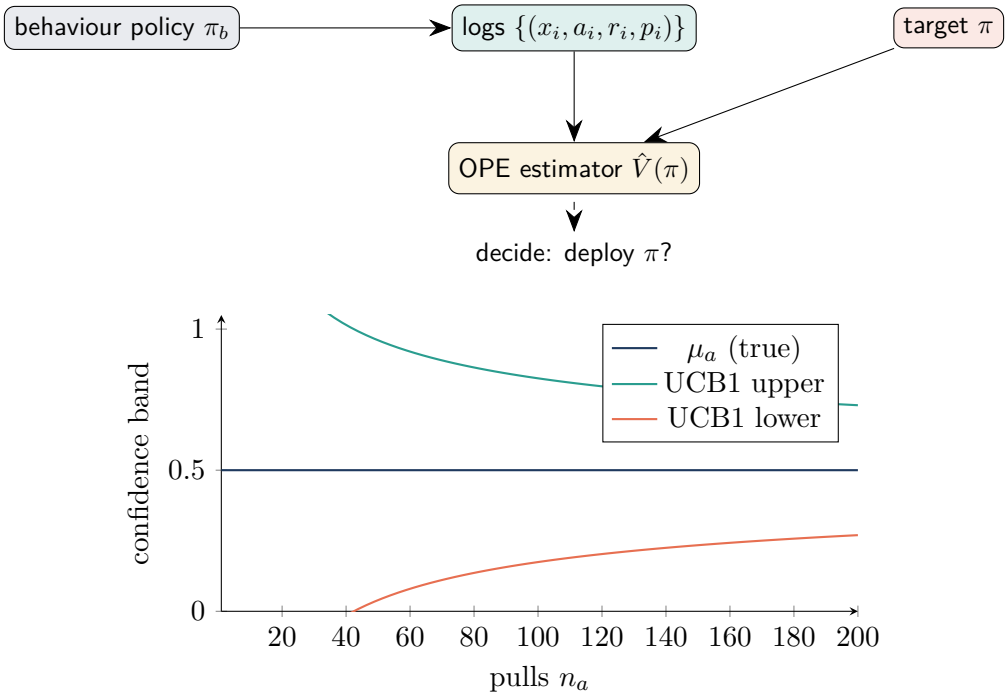
A diagnostic for whether OPE estimates are trustworthy:

$$\mathrm{ESS} \triangleq \frac{(\sum_i w_i)^2}{\sum_i w_i^2}.$$

$\mathrm{ESS} \in [1, n]$. $\mathrm{ESS} \ll n$ indicates that a tiny fraction of log entries carries most of the weight — the estimator is effectively relying on a handful of samples. Common production thresholds: refuse to report an OPE number if $\mathrm{ESS} < 100$ or $\mathrm{ESS}/n < 0.05$.

Intuition. ESS is the bandit analogue of the rank of a design matrix in linear regression: it tells you how much information the data actually contains about the quantity you are estimating. A 1-million-row log with ESS = 50 is a 50-row dataset wearing a costume.

2.9 Visual intuition



Common pitfall

Positivity is not a luxury. If $\pi_b(a \mid x) = 0$ for some (x, a) the target might assign, IPS and DR are biased and SNIPS is silently wrong. Use ε-greedy or stochastic propensity floors at logging time.

3 Worked numerical examples and pitfalls

3.1 Example: ESS collapse

Let $K = 100$ and $\varepsilon = 0.05$. Suppose the target always chooses an arm that is *not* the logger's greedy arm. Its logging probability is $p = \varepsilon/K = 0.0005$. A logged target action has weight 2000; every other logged action has target probability zero and weight zero. With $n = 10\,000$, the expected number of nonzero rows is $np = 5$. If the realised count is $m > 0$, then ESS $= (2000m)^2/(2000^2 m) = m$; at $m = 0$ both SNIPS and ESS have zero denominators. When logger and target agree on the greedy arm, the nonzero weight is instead $1/0.9505$ and almost all rows contribute.

3.2 Example: clipping changes the answer

A two-action logger chooses B with probability 0.01; the target always chooses B, whose deterministic reward is 1. Then $w = 100$ on B and 0 otherwise. Unclipped IPS has expectation 1; clipping at 10 gives expectation 0.1. This valid probability model satisfies $\mathbb{E}_b[w] = 1$ before clipping. Report lost weight mass and sensitivity to the cap; a fixed fraction of clipped rows is not a universal validity test.

3.3 Example: positivity violation

Suppose your logger is deterministic on lead category "commercial motor" (always sends to broker 3). A target policy wants to send commercial-motor leads to broker 4. The importance weight for those samples is zero; IPS returns zero on them; the estimate is biased downward. The fix is structural: either randomise the logger (ε-greedy with $\varepsilon > 0$) or restrict the target to actions the logger actually explores.

3.4 Pitfalls compendium

Common pitfall

Confusing π_b and target π. The propensity is the behaviour policy's probability of taking the logged action, not the target's. The sanity-check is that the unweighted average of importance weights converges to 1.

Common pitfall

Counting clipped samples as un-clipped. A common production sin: "we clipped 30% of weights and the estimator looks fine". It is not fine; clipping changes the estimand; with rewards in $[0, 1]$, its downward expectation bias is bounded by $\mathbb{E}_b[(w - c)_+]$. Report this lost mass and uncertainty rather than assume clipping is harmless.

Common pitfall

OPE across policy versions. Different logger versions can be combined if each row records its actual conditional action propensity and the target estimand is consistent. Adaptive logging changes the uncertainty analysis; distribution shift or missing version-specific probabilities requires additional treatment. A version boundary alone does not invalidate IPS.

4 Algorithms

Algorithm 1: LinUCB (disjoint arms)

Input: K, d, exploration radius α, ridge λ.
Init: $A_a \leftarrow \lambda I_d$, $b_a \leftarrow 0$ for all a.
for $t = 1, 2, \dots$ **do**
  observe x_t
  for $a \in [K]$ **do**
    $\hat{\theta}_a \leftarrow A_a^{-1} b_a$
    $p_a \leftarrow \hat{\theta}_a^\top x_t + \alpha \sqrt{x_t^\top A_a^{-1} x_t}$
  pick $A_t = \arg\max_a p_a$
  observe r_t
  $A_{A_t} \leftarrow A_{A_t} + x_t x_t^\top$
  $b_{A_t} \leftarrow b_{A_t} + r_t x_t$

Algorithm 2: LinTS (Gaussian posterior sampling)

Input: K, d, posterior scale v, ridge λ.
Init as LinUCB.
for $t = 1, 2, \dots$ **do**
  observe x_t
  for $a \in [K]$ **do**
    sample $\tilde{\theta}_a \sim \mathcal{N}(A_a^{-1} b_a,\ v^2 A_a^{-1})$
    $p_a \leftarrow \tilde{\theta}_a^\top x_t$
  pick $A_t = \arg\max_a p_a$, observe r_t, update A_{A_t}, b_{A_t} as in LinUCB.

Algorithm 3: Doubly Robust OPE

Input: log $\mathcal{D}$, target π, reward model $\hat{q}$.
for $(x_i, a_i, r_i, p_i) \in \mathcal{D}$ **do**
  $w_i \leftarrow \pi(a_i \mid x_i) \,/\, p_i$
  $v_i \leftarrow \sum_a \pi(a \mid x_i) \hat{q}(x_i, a) + w_i (r_i - \hat{q}(x_i, a_i))$
return $\frac{1}{n} \sum_i v_i$.

Example 4.0.1 (Worked numerical example)**.** Suppose $K = 2$, logger $\pi_b = (0.7, 0.3)$ (context-free for clarity), target $\pi = (0.2, 0.8)$. A log of $n = 4$ samples:

i	a_i	r_i	p_i	w_i	$w_i r_i$
1	1	1	0.7	0.286	0.286
2	2	0	0.3	2.667	0.000
3	2	1	0.3	2.667	2.667
4	1	0	0.7	0.286	0.000

$\hat{V}_{\text{IPS}} = (0.286 + 0 + 2.667 + 0)/4 = 0.738$. The single sample $i = 3$ with $w_i = 2.67$ dominates — this is the classic IPS variance explosion when the logger places little mass on the target's preferred action.

Complexity. LinUCB per round: $O(Kd^2)$ for the score loop plus $O(d^2)$ for the rank-1 update (Sherman–Morrison gives this in closed form if you maintain A_a^{-1}). LinTS adds a $O(d^3)$ Cholesky per arm per round; cache it.

Common pitfall

Estimated propensities and policy overfitting are different. Estimating a logger model can introduce misspecification and estimation error; it does not inherently leak rewards. Propensity overfitting in counterfactual policy learning refers to exploiting the weighted training objective. Prefer exact logged probabilities when available, and use held-out policy evaluation and appropriate nuisance-model splitting.

5 Implementation walkthrough

The mini-project lives in `project/`. The environment is `SyntheticLeadBandit`: $K = 6$ brokers, $d = 8$ lead features, hidden linear preferences with broker-specific feature specialisation and a quality gradient (broker 0 best on average, broker 5 worst).

```
class SyntheticLeadBandit:
    d: int = 8
    K: int = 6
    # theta[a] in R^d encodes broker a's hidden preferences
    def mean_reward(self, x, a):
        return sigmoid(x @ self.theta[a])
    def step(self, x, a):
        return rng.binomial(1, self.mean_reward(x, a))
```

LinUCB is one screen of code:

```
class LinUCB:
    def __init__(self, K, d, alpha=1.0, lam=1.0):
        self.A = [lam*np.eye(d) for _ in range(K)]
        self.b = [np.zeros(d)    for _ in range(K)]
        self.alpha = alpha
    def act(self, x):
        scores = []
        for a in range(self.K):
            Ainv_x = np.linalg.solve(self.A[a], x)
            mu = self.b[a] @ Ainv_x
            ucb = self.alpha * np.sqrt(max(x @ Ainv_x, 0))
            scores.append(mu + ucb)
        return int(np.argmax(scores))
    def update(self, x, a, r):
        self.A[a] += np.outer(x, x); self.b[a] += r * x
```

The offline logger is ε-greedy on its own ridge model, and *records the propensity at decision time*:

```
def act(self, x):
    p = self.propensities(x)
    a = rng.choice(self.K, p=p)
    return a, p[a]    # propensity is observed, not re-estimated
```

The DR arithmetic uses validated rows and finite, correctly shaped reward predictions. The reference nuisance model is fitted on an independent uniform sample with the same latent environment; see `independent_logged_sample` in `project/ope.py`. Fitting on independent data separates evaluation from nuisance fitting; it does not remove model misspecification.

```
def doubly_robust(log, pi, qhat):
    s = 0.0
    for x, a, r, pix, w in _checked_entries(log, pi):
```

```
        q = _predictions(qhat, x, pix)
        baseline = float(pix @ q)
        s += baseline + w * (r - float(q[a]))
    return s / len(log)
```

The full code is in `project/agents.py`, `project/ope.py`, `project/train.py`. The reference `train()` function (target: LinTS regret $< 0.6T$ at $T = 2000$) runs in about 10 seconds on CPU.

6 Production pipeline: synthetic insurance in detail

This chapter walks through the engineering of the synthetic insurance lead-broker allocation pipeline at the level of detail an SRE / ML-engineer can build against. The cascade has the same structure as the ATS pipeline in Week 12 (retrieval → rerank → explore → audit → calibrate), but with degenerate retrieval (the broker panel is small).

6.1 Data contract

Every lead arriving at the allocator carries a typed payload:

- `lead_id`: a UUID for log joining.
- `features`: postcode band, vehicle group, prior-claims count, channel (direct vs. aggregator), age band, NCD years, quote channel, marketing source. Each is a categorical or numeric field with a documented type.
- `ingest_timestamp`: when the lead arrived; needed for delay-distribution modelling.
- `deduplication_hash`: catch leads submitted multiple times across channels; treat duplicates carefully (do not log multiple allocations).

6.2 Allocation decision

The allocator runs LinTS over the panel of K brokers. The decision step:

1. Compute $\tilde{\theta}_a \sim \mathcal{N}(\hat{\theta}_a, v^2 A_a^{-1})$ per broker.
2. Compute $\tilde{s}_a = x^\top \tilde{\theta}_a$ per broker.
3. Apply hard constraints: drop brokers who are offline, at quota, or have rejected this lead category before.
4. Apply exposure budgets (regulator-facing fairness floor).
5. Sample one broker proportional to $\text{softmax}(\tilde{s}_a/\tau)$; or take argmax with ε-greedy exploration.

For pure Thompson argmax sampling, the marginal action propensity integrates over all posterior draws; one sampled score vector does not supply it. A stochastic wrapper conditional on a logged posterior draw can expose its own conditional probability, but OPE must explicitly use that enlarged context and the same draw distribution. The course logger uses an explicit epsilon-greedy mixture with directly known probabilities. Any mask or fairness post-processing must be included in the final policy.

6.3 Logging

The proposed decision log records context and action fields $(x, a, \pi_b(a \mid x))$, posterior diagnostics $(\tilde{\theta}_a, v, A_a^{-1})$, and version/timestamp fields `policy_version`, `model_hash`, `ingest_ts`, and `decision_ts`. A Kafka topic partitioned by hour is one storage option. The diagnostic fields ($\tilde{\theta}_a$, v, A_a^{-1}) are needed only for forensics; the critical fields for OPE are $(x, a, r, \pi_b(a \mid x)$, policy_version).

6.4 Delayed outcome

A lead either binds (within 1–90 days) or expires. The reward r is right-censored: at evaluation time, some leads' fate is still unknown. The logger writes a partial record at decision time; an outcome-merger service writes the binding event (or expiry) when it arrives, joined on `lead_id`.

Define the reward window first, then evaluate cohorts whose outcome window has matured or use a justified censoring model. Waiting only the median delay leaves roughly half of delayed outcomes unresolved. A survival model requires correct censoring and outcome assumptions; it does not make every recent-cohort estimate automatically unbiased.

6.5 Audit

- *Daily*: propensity-overlap monitor (ESS / n on the OPE replay), calibration-drift monitor (sliding-window ECE).
- *Weekly*: DR-OPE on the next-candidate policy.
- *Monthly*: fairness audit (broker exposure by lead category; the synthetic insurance version of demographic-parity is exposure-by-broker).
- *Quarterly*: delayed-feedback model recalibration (Chapelle survival head).

6.6 Kill-switch

The production policy is gated behind a circuit breaker: if the live bind rate drops below 80% of the previous week's average, the allocator reverts to the previous policy version within seconds. The kill-switch prevents a bad LinTS posterior from cascading into a revenue incident.

7 Empirical results

7.1 Statistics bridge: A defined estimate can still have almost no observed overlap

Example 7.1.1 (Original statistical teaching fixture)**.** Ten independent logged rows contain exactly one target-action observation. The logger probability for that action is 0.1, the deterministic target always chooses it, and its observed reward is 1.

Weights are $(10, 0, \ldots, 0)$. IPS is $10/10 = 1$, SNIPS is $10/10 = 1$, and weight ESS is $10^2/10^2 = 1$. An ordinary ten-row bootstrap omits the sole contributing row with probability $(9/10)^{10} \approx 0.34868$, making that resample SNIPS undefined.

Agreement of two estimates is not evidence of precision. Population support is positive in the fixture, while observed overlap is only one row. More bootstrap draws do not create another target-action outcome; silently discarding undefined resamples changes the reported analysis.

Check a changed assumption. Clip the weight to 5 and calculate IPS. What did the numerical stabilization change? Which uncertainty is absent if a reward model is frozen during resampling?

Check. Clipped IPS is 0.5. Clipping changes the estimator and its bias, rather than adding information. A frozen-model bootstrap omits refitting and model-selection variability; it answers a conditional question.

Statistical foundation. Siegrist § § 4.1, 7.1, PDF pp.439, 896; Ross § 15.3, PDF p.683. Section source: https://stats.libretexts.org/@go/page/10189. The course statistics companion records the supplied editions and one-based PDF pages. This fixture is not a newly measured project result.

We train LinUCB and LinTS for $T = 2000$ rounds on the synthetic insurance bandit (Figure 7.1). The reward means are logistic and feature draws are not the bounded linear model of the theorem. These finite synthetic curves illustrate behaviour, not a proof of that regret rate.

Figure 7.2 holds one synthetic reward model and softmax target fixed. Each repetition generates a new adaptive logging history and fits a new reward nuisance model on a separate independent uniform log from the same environment. The independent repetition is the whole log and its fit, not a row from the changing logger. The bars' SD is descriptive repeated-experiment spread. It is not a confidence interval, and it does not measure variation across different latent reward models. SNIPS bias has no universal sign; DR has no unconditional variance ordering.

The shared reference integrates the simulator's known target-weighted reward means over 50,000 independent fresh contexts. It averages out action and reward noise analytically, but retains context Monte Carlo error. Its reported MC SE describes numerical integration error, separately from estimator variability. The reference is not exact ground truth. Raw estimates, seeds and reference metadata are saved alongside the figures in `fig_ope_bias.json` and `fig_bias_variance.json`.

Figure 7.3 shows the importance-weight distribution. Under the oracle-greedy target, weights may approach K/ε, but cannot exceed this bound when the epsilon-greedy floor is correctly logged and target probabilities are at most one. Large weights motivate overlap diagnostics.

Figure 7.4 uses its own fixed oracle-greedy target and corresponding shared MC reference on twenty independently collected adaptive logs, with independent nuisance fits. Each threshold

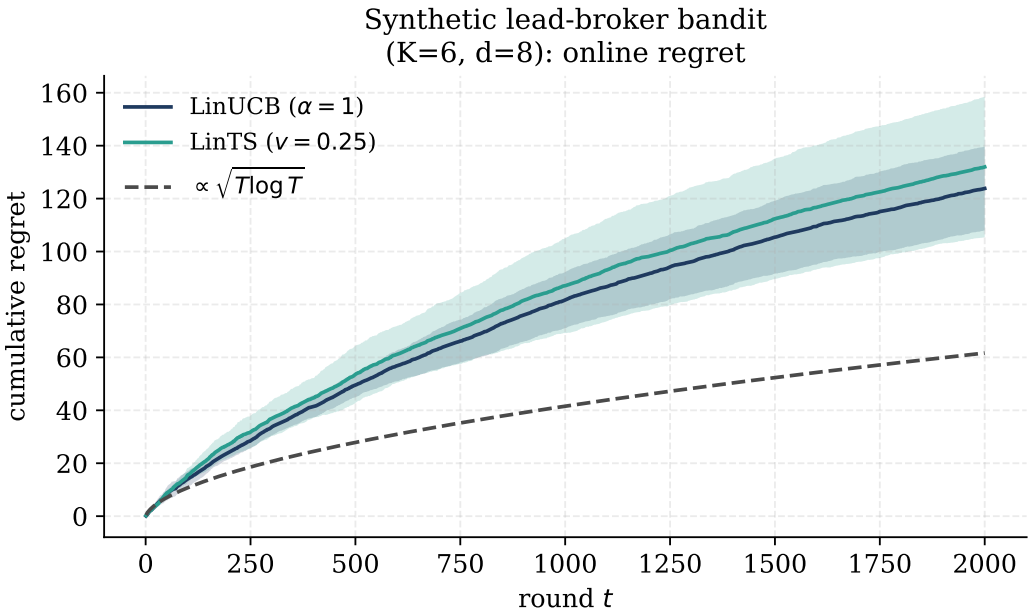


Figure 7.1: Cumulative pseudo-regret on the synthetic insurance lead-broker bandit, mean ± empirical SD across four seeded environment instances. This spread includes changing latent instances and sampled histories, rather than repeated estimation of one fixed policy value. The dashed $\sqrt{T \log T}$ curve is a visual reference shape, not a verified growth rate or confidence bound for these logistic-reward implementations.

is evaluated on the same repetitions. Write $e_r(\tau) = \widehat{V}_r(\tau) - V_{\text{ref}}$ and $\bar{e}(\tau) = R^{-1} \sum_r e_r(\tau)$. The finite empirical decomposition is

$$\underbrace{\frac{1}{R}\sum_r e_r^2}_{\text{empirical MSE}} = \underbrace{\bar{e}^2}_{\text{squared empirical mean error}} + \underbrace{\frac{1}{R}\sum_r (e_r - \bar{e})^2}_{\text{empirical variance, denominator } R} .$$

All three quantities have squared reward units. The squared mean error is relative to the shared MC reference, rather than a known population squared bias. An $R-1$ sample variance would not satisfy this particular identity. For errors $(-0.1, 0.1, 0.2)$, the terms are $1/50 = 1/225 + 7/450 = 0.02$.

Small τ removes some weighted residual corrections and can introduce model-dependent bias; neither variance nor MSE has a universal monotone ordering. When τ includes every observed weight, SWITCH equals DR on that log. Mean-correctness arguments for DR require the stated support and conditional propensity or reward-model assumptions; they do not make a finite repeated-log average equal the numerical reference. Its MC error shifts all errors together, so it contributes uncertainty to the empirical squared-error summaries.

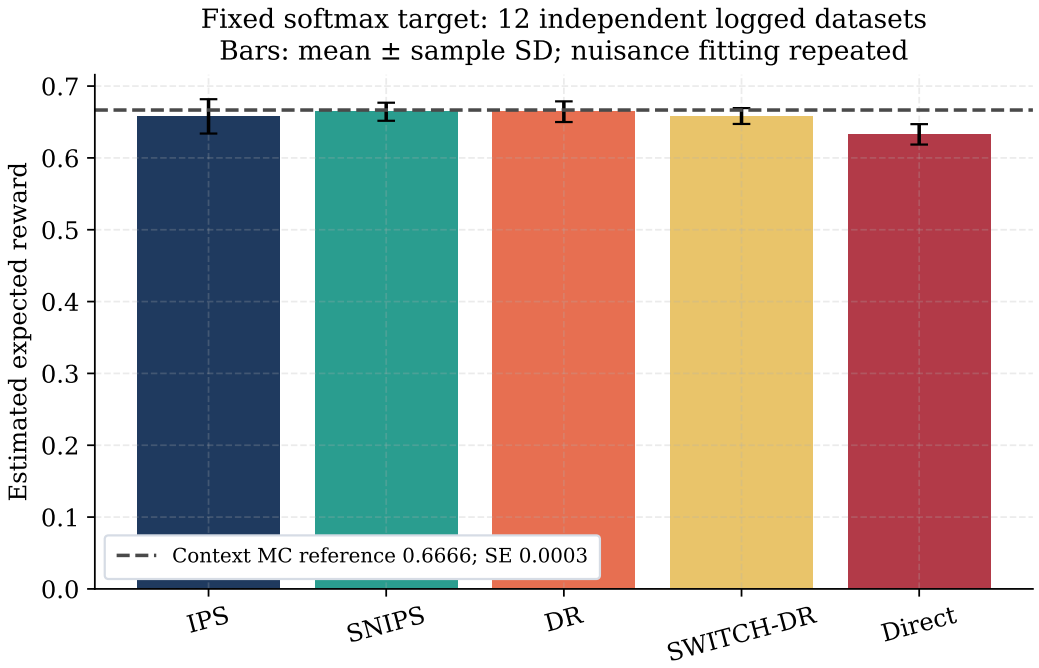


Figure 7.2: OPE estimates for one fixed softmax target and fixed synthetic reward model: mean ± sample SD (denominator 11) over twelve independent whole-log and nuisance-fit repetitions, with 4000 evaluation rows and 2000 independent uniform fitting rows per repetition. The dashed line is one shared 50,000-context MC reference; its MC SE is reported separately in the figure and saved metadata. These descriptive bars are not confidence intervals.

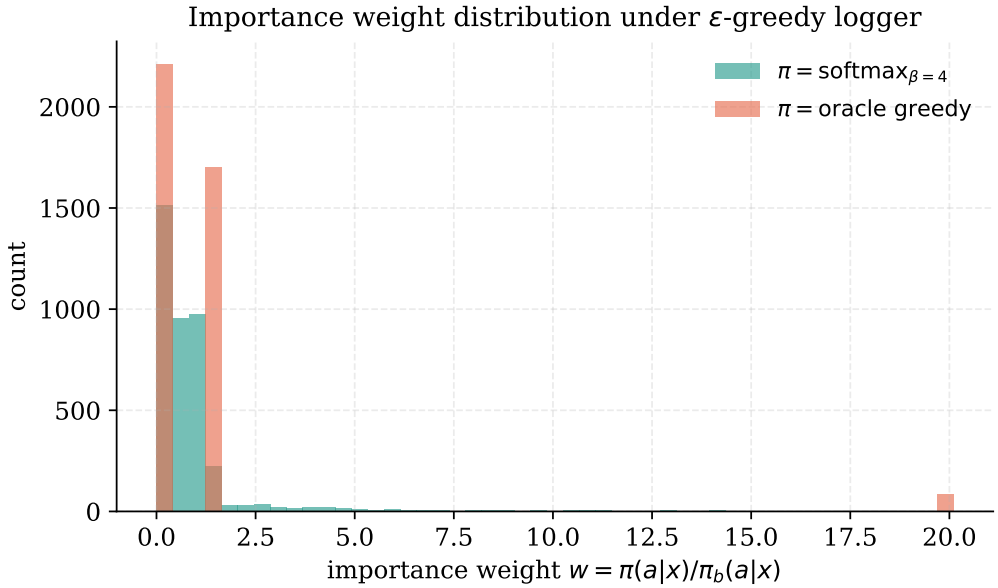


Figure 7.3: Observed importance weights in one 4000-row adaptive log, with six actions and exploration $\varepsilon = 0.3$. The minimum logging propensity is $\varepsilon/K = 0.05$, so the oracle-greedy target's weights can reach 20. The histogram describes observed overlap; it neither proves population support nor supplies an independent-row confidence interval.

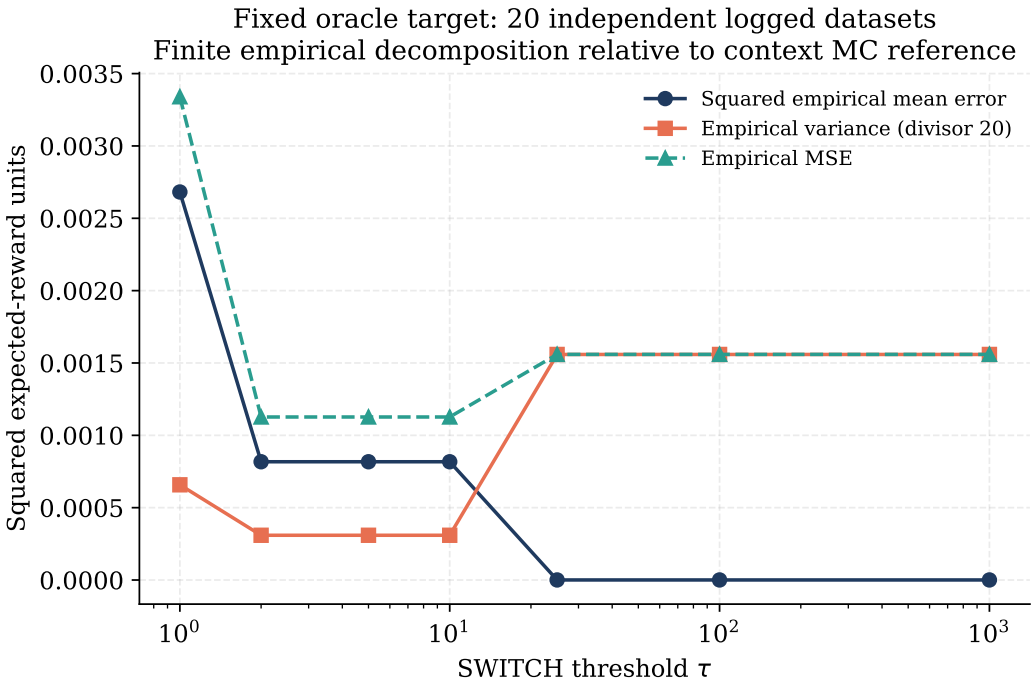


Figure 7.4: SWITCH-DR squared mean error, empirical variance (denominator 20), and MSE relative to one shared MC reference, across twenty independent whole-log and nuisance-fit repetitions for one fixed oracle target and reward model. All ordinates have squared reward units and satisfy the finite empirical decomposition. The comparison is conditional on this model, target, logger, fitting protocol and numerical reference; it establishes no universal threshold ranking or exact population bias.

8 Safety angle

Safety angle

Insurance lead routing is a *high-stakes, delayed-feedback* domain. Concrete failure modes:
F1 — Unrestricted exploration. A reward-seeking explorer may choose actions with unacceptable costs. Specify admissible actions and an explicit baseline constraint before evaluating a conservative method. Conservative Bandits (Wu, Shariff, Lattimore and Szepesvári, ICML 2016) controls performance relative to a baseline under its stochastic-bandit assumptions; it is not merely a per-arm epsilon cap or a universal high-confidence deployment test. Source: https://proceedings.mlr.press/v48/wu16.html.
F2 — IPS with missing propensities. A common production sin: the production policy is changed by an engineer, the propensity logger is not updated, and OPE silently becomes wrong. *Mitigation:* log policy version hashes alongside the propensity; verify each version's conditional propensity and account for adaptive logging in uncertainty estimates.
F3 — Delayed feedback bias. A lead binds in 1–90 days. If you evaluate today, recent leads are systematically under-credited. The estimator is biased *against* new policies (which only have recent data). *Mitigation:* use fully matured outcome windows, or a validated censoring/survival model with explicit assumptions (Chapelle 2014; Vernade et al. 2017).
F4 — Overlap (positivity) failures. A new broker (cold-start) has zero historical mass on most leads. Any target policy that prefers them is un-evaluable by IPS / DR. *Mitigation:* hierarchical Thompson priors (Kveton et al. 2021); explicit cold-start exploration budgets; honest reporting of ESS (effective sample size).
F5 — Propensity overfitting. Policy selection on the same weighted log can exploit its noise. Estimated propensities pose a separate model-error risk. *Mitigation:* log propensities at decision time.
Two cites: Bottou et al., *Counterfactual Reasoning and Learning Systems*, JMLR 2013; Joachims, Swaminathan, de Rijke, *Deep Learning with Logged Bandit Feedback*, ICLR 2018 (BanditNet).

9 Matching application

Matching application

The synthetic insurance lead-broker problem is the canonical applied bandit. The mapping:

Bandit primitive	synthetic insurance analogue
context x	lead features (postcode, vehicle, age, channel, prior claims)
arm a	broker (one of K panel members)
reward r	bind / no-bind (binary)
propensity $\pi_b(a\|x)$	probability the production allocator chose broker a for lead x
target π	candidate policy (re-weighted LinTS, or a fairness-constrained variant)
delayed feedback	1–90 days from quote to bind

Engineering implications:

1. Log $(x, a, r, p,$ policy_version_hash$)$ *at decision time*.
2. Train a calibrated $\hat{q}(x, a)$ (Beta-calibration on a holdout split).
3. Run weekly DR + SWITCH evaluations against the next-candidate policy.
4. Gate deploys behind a conservative bandit bound (Wu et al. 2016).
5. Run a delayed-feedback adjustment (survival head, Chapelle 2014).

A similar mapping works for ATS candidate–job ranking (Week 12), but position bias and slate structure add a layer of correction.

10 Exercises

See `workbook.ipynb` for full statements, hints, and graded tests. Summary:

- **E1** ($\star$) Compute the UCB1 bonus for $t = 100$, $n_a = 10$.
- **E2** ($\star$) Implement the KL-UCB upper-confidence solver.
- **E3** ($\star\star$) Bernoulli Thompson Sampling on a 3-armed bandit; compare finite-horizon curves across independent runs with the theoretical assumptions. Explain why the plot cannot confirm an asymptotic $O(\log T)$ rate.
- **E4** ($\star\star$) Train LinUCB on the synthetic insurance bandit; report cumulative regret.
- **E5** ($\star\star$) Collect ε-greedy logs; compute IPS and SNIPS for a uniform target.
- **E6** ($\star\star$) Doubly Robust against the softmax target; compare to a fresh-context MC reference and report its MC SE separately.
- **E7** ($\star$) Effective sample size of importance weights.
- **E8** ($\star\star$) SWITCH-DR; vary τ on independent repetitions of a fixed target/model and calculate empirical squared mean error, variance with denominator R, and MSE.
- **E9** ($\star\star\star$, *open-ended*, *synthetic insurance objective*) For three counterfactual policies (uniform, softmax-true, oracle-greedy) compare SNIPS, DR, and SWITCH-DR estimates against their own fresh-context MC references. Hold each target/model fixed across independent whole-log repetitions and report reference MC error separately. Write a one-page bounded recommendation memo to the synthetic insurance allocation team.

11 Further reading

The `extended_reading.pdf` document contains annotated summaries of five foundational papers: Auer et al. 2002 (UCB1), Agrawal & Goyal 2013 (LinTS), Li et al. 2010 (LinUCB), Dudík et al. 2011 (DR), Joachims et al. 2018 (BanditNet). For broader context, Lattimore & Szepesvári, *Bandit Algorithms* (2020), is the textbook; Bietti, Agarwal, Langford, *A Contextual Bandit Bake-off* (JMLR 2021) is the empirical state of the art; and Saito & Joachims (ICML 2022) introduce MIPS for large-action OPE — the natural next step for synthetic insurance / ATS when K grows from 6 to 6000.

Sourced from `research/bandits_matching_sota.md` sections 1–3 (stochastic bandits, contextual bandits, OPE) and `research/safety_rl_sota.md` (conservative bandits, batch RL safety).